TASK SPECIFICATION: P1-02 PHASE P1 — FOUNDATIONS & DEVOPS INFRASTRUCTURE

Frontend Application Scaffold (React 19 + TypeScript Strict)

Git Branch: `feature/P1-02-frontend-scaffold` · Target Pull Request: PR (feature/P1-02-frontend-scaffold)

Core Objective & Technical Scope:
Initialize high-performance React 19 + TypeScript production application in frontend/ with strict typechecking, ESLint 9, Prettier, and Vitest testing suite.

Priority Level	High	Complexity Rating	`M`
Execution Status	Done ✅	Target PR / Commit	`PR (feature/P1-02-frontend-scaffold)`
Upstream Dependencies	P1-01, ADR-0002		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Bootstrap React 19 + TypeScript project using Vite 6.
2. Configure strict TypeScript compiler options and ESLint Flat Config.
3. Configure Vitest testing environment with React Testing Library and jsdom.
4. Establish enterprise folder architecture (app/, features/, shared/, styles/).
5. Implement standardized scripts: dev, build, lint, format, typecheck, test.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `frontend/package.json`
- `frontend/tsconfig.json`
- `frontend/vite.config.ts`
- `frontend/eslint.config.js`
- `frontend/src/**`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- All npm scripts (build, lint, typecheck, test) execute with zero warnings or errors.
- Unit test suite passes with jsdom environment configured.

VERIFICATION & TEST COVERAGE STRATEGY

- npm run lint
- npm run typecheck
- npm run test
- npm run build